

No SAE J1939 Communication Remote Panel

Symptoms

- No Society of Automotive Engineers (SAE) J1939 communication to the control panel
- No SAE J1939 communication with the remote panel display panel
- Engine room panel has SAE J1939 communication.

How To Use This Tree

This symptom tree can be used to troubleshoot a malfunction. Step 1 troubleshooting. Step 2 will ask a series of questions and will provide a list of troubleshooting steps to perform, depending on the symptom.

Shoptalk

Start by checking the terminating resistor. The terminating resistor is located on the QSM11 and QSB7-DM CM850 wiring diagrams on the engine wiring harness.

The SAE J1939 data link provides information to the display in the remote panel.

The SAE J1939 data link provides the following parameters:

- Engine fault codes
- Engine parameters monitored by the engine control module (ECM).

Troubleshooting Summary

STEPS		SPECIFICATIONS
STEP 1.	Check the customer interface box (C.I.B.).	
	STEP 1A. Check the control panel display for faults.	Control panel indicates fault(s)?
	STEP 1A-1. Check the control panel power SUPPLY wire for voltage +24 VDC.	Less than +24 VDC?
STEP 2.	Check the SAE J1939 data link signal.	

STEPS		SPECIFICATIONS
	STEP 2A. Check the SAE J1939 data link communication on the engine.	Communication established?
	STEP 2B. Check the SAE J1939 data link communication at the remote panel.	Communication established?
STEP 3.	Check the remote panel wiring.	
	STEP 3A. Check the SAE J1939 SUPPLY wire for an open circuit.	Less than 10 ohms?
	STEP 3B. Check the SAE J1939 RETURN wire for an open circuit.	Less than 10 ohms?
	STEP 3C. Check the SAE J1939 data link shield wire for an open circuit.	Less than 10 ohms?
STEP 4.	Check the remote panel wiring.	
	STEP 4A. Check the SAE J1939 data link SUPPLY, RETURN, and shield wires for a wire-to-wire short circuit.	Less than 10 ohms?
	STEP 4B. Check the SAE J1939 data link SUPPLY, RETURN, and shield wires for a short circuit to ground.	Less than 10 ohms?

STEP 1. Check the customer interface box (C.I.B.).

STEP 1A. Check the control panel display for faults.

Conditions :

- Locate the control panel display.
- Connect the recommended Cummins® electronic service tool.

Action	Specification/Repair	Next Step
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Check the control panel display for indication of faults.	Control panel indicated fault(s)? YES	2A
	Control panel indicated fault(s)? NO	1A-1

STEP 1A-1. Check the control panel power SUPPLY wire for voltage +24 VDC.

Conditions :

- Open the C.I.B.

Action	Specification/Repair	Next Step
Check the voltage at the battery 1 voltage (switched power) at the control panel. • Place one test on the battery 1 voltage SUPPLY wire at the control panel. • Place the other test lead on the panel ground. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than +24 VDC? YES Repair : Check the batteries. Refer to the original equipment manufacturer (OEM) service manual.	Repair complete
	Less than +24 VDC? NO	2A

STEP 2. Check the SAE J1939 data link signal.

STEP 2A. Check the SAE J1939 data link supply communication on the engine.

Conditions :

- Locate the engine wiring harness from the ECM.
- Connect the recommended Cummins® electronic service tool.

Action	Specification/Repair	Next Step
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Check the SAE J1939 data link supply communication on the engine. Use the recommended Cummins® electronic service tool to establish communication.	Communication established? YES	2B
	Communication established? NO Repair : Troubleshoot the appropriate fault code. Reference Marine Auxiliary QSB7-DM CM850 Fault Code Troubleshooting Manual, Bulletin 4325972, Section TF; or ISM and QSM 11 Electronic Control System Troubleshooting and Repair Manual, Bulletin 3666266, Section TF; or X15 CM2350 X125M Fault Code Troubleshooting Manual, Bulletin 5504346, Section TF.	Repair complete

STEP 2B. Check the SAE J1939 data link communication at the remote panel.

Conditions : - Open the C.I.B. - Connect the recommended Cummins® electronic service tool.		
Action	Specification/Repair	Next Step
Check the SAE J1939 data link communication at the remote panel. Use the recommended Cummins® electronic service tool to establish communication.	Communication established? YES	Repair complete
	Communication established? NO	3A

STEP 3. Check the remote panel wiring.

STEP 3A. Check the SAE J1939 SUPPLY wire for an open circuit.

Conditions :

- Open the C.I.B.
- Locate the remote panel display.
- Disconnect the SAE J1939 SUPPLY wire at the control panel and service port connector.
- Disconnect the C1 connector.

Action	Specification/Repair	Next Step
Check the SAE J1939 SUPPLY wire for an open circuit. • Place one test lead on the SAE J1939 data link SUPPLY wire at the control panel. • Place the other test lead on the SUPPLY wire at the SAE J1939 data link service port connector. • Place one test lead on the SAE J1939 data link SUPPLY wire at the control panel. • Place the other test lead on the SAE J1939 data link SUPPLY wire at the C1 connector.	Less than 10 ohms? YES	3B
	Less than 10 ohms? NO Repair : Replace the wire. Refer to Procedure 015-023 in Section 15. (/qs3/pubsys2/xml/en/procedures/300/300-015-023.html)	Repair complete

STEP 3B. Check the SAE J1939 RETURN wire for an open circuit.

Conditions :

- Open the C.I.B.
- Locate the remote panel display.
- Disconnect the SAE J1939 RETURN wire at the control panel and service port connector.
- Disconnect the C1 connector.

Action	Specification/Repair	Next Step
Check the SAE J1939 RETURN wire for an open circuit. • Place one test lead on the SAE J1939 data link RETURN wire at the control panel. • Place the other test lead on the RETURN wire at the SAE J1939 data link service port connector. • Place one test lead on the SAE J1939 data link RETURN wire at the control panel. • Place the other test lead on the SAE J1939 data link RETURN wire at the C1 connector.	Less than 10 ohms? YES	3C
	Less than 10 ohms? NO Repair : Replace the wire. Refer to Procedure 015-023 in Section 15. (/qs3/pubsys2/xml/en/procedures/300/300-015-023.html)	Repair complete

STEP 3C. Check the SAE J1939 shield wire for an open circuit.**Conditions :**

- Open the C.I.B.
- Locate the remote panel display.
- Disconnect the SAE J1939 shield wire at the control panel and service port connector.

Action	Specification/Repair	Next Step
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Check the SAE J1939 shield wire for an open circuit. - Place one test lead on the SAE J1939 data link shield wire at the control panel. - Place the other test lead on the shield wire at the SAE J1939 data link service port connector.	Less than 10 ohms? YES	4A
	Less than 10 ohms? NO Repair : Replace the wire. Refer to Procedure 015-023 in Section 15. (/qs3/pubsys2/xml/en/procedures/300/300-015-023.html)	Repair complete

STEP 4. Check the remote panel wiring.

STEP 4A. Check the SAE J1939 data link SUPPY, RETURN, ans shield wires for a wire-to-wire short circuit.

Conditions : - Open the C.I.B. - Locate the remote panel display.		
Action	Specification/Repair	Next Step

Check the SAE J1939 data link SUPPLY, RETURN, and shield wires for a wire-to-wire short circuit. - Place one test lead on the SAE J1939 data link SUPPLY wire at the control panel. - Place the other test lead on all other wires at the control panel. - Place one test lead on the SAE J1939 data link RETURN wire at the control panel. - Place the other test lead on all other wires at the control panel. - Place one test lead on the SAE J1939 data link shield wire at the control panel. - Place the other test lead on all other wires at the control panel. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? YES Repair : Replace the control panel. Contact a Cummins® Authorized Repair Location.	Repair complete
	Less than 10 ohms? NO	4A

STEP 4B. Check the SAE J1939 data link SUPPLY, RETURN, and shield wires for a short circuit to ground.

Conditions :

- Open the C.I.B.
- Locate the remote panel display.

Action	Specification/Repair	Next Step
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Check the SAE J1939 data link SUPPLY, RETURN, and shield wires for a short circuit to ground. • Place one test lead on the SAE J1939 data link SUPPLY wire at the control panel. • Place the other test lead on all other wires at the control panel. • Place one test lead on the SAE J1939 data link RETURN wire at the control panel. • Place the other test lead on all other wires at the control panel. • Place one test lead on the SAE J1939 data link shield wire at the control panel. • Place the other test lead on all other wires at the control panel. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? YES Repair : Replace the control panel. Contact a Cummins® Authorized Repair Location.	Repair complete
	Less than 10 ohms? NO	Contact a Cummins® Authorized Repair Location.